

Tachycardia (Narrow Complex with Pulses)

History

- Is the patient tachycardic (HR > 100)?
- Is the patient symptomatic?
- If symptomatic, when did the symptoms start?
- Did patient have recent trauma? Toxic ingestion?
- Does patient have suspected infection? Fever? Heat exposure? Dehydration?
- Patient's medical history – Cardiac history? History of arrhythmia? History of prolonged QT syndrome? Renal failure? Diabetes? Lung disease?
- Medications – Many medications can lead to tachyarrhythmias

Signs and Symptoms of Hemodynamically Significant Tachycardia

- Hypotension
- Syncope / Dizziness / Lightheadedness
- Chest pain
- Shortness of breath / Difficulty breathing
- Signs of pulmonary edema (crackles, rales, hypoxia)
- Altered mental status / Agitation
- Diaphoresis
- Cyanosis or decreased capillary refill

EMT/AEMT

- Follow General Medical Care Guideline
- Follow Adult Resuscitation Guideline, as indicated

EMT/AEMT - Expanded Scope

- **NS/LR** 500ml IO for SBP < 90 or signs/symptoms of shock

Paramedic

- 12 – Lead EKG
- Advanced airway as indicated
- **NS/LR** 500-1000 mL IV/IO bolus for SBP ≤ 90 or signs/symptoms of shock

Sinus Tachycardia or Heart Rate < 150

- 12 – Lead interpretation

Heart Rate > 150 and SBP ≤ 90

- Immediate **Synchronized Cardioversion** – 100 J → 150 J → 200 J biphasic
 - Reassess rhythm between cardioversions
 - If patient converts, perform or repeat 12 – Lead
- If no cardioversion, ensure IV fluid of at least 1000mL --> reassess blood pressure --> follow appropriate guideline --> continue IV fluid during resuscitation

Heart Rate > 150 and SBP > 90 with Regular Rhythm (SVT)

- Attempt Vagal Maneuvers / Modified Valsalva
- **Adenosine** 12 mg rapid IV push
 - Repeat x 1
 - If no conversion repeat 12 – Lead
- If no conversion, **Metoprolol** 5mg slow IVP over 2 mins may repeat in 5 mins

Heart Rate > 150 and SBP > 90 with Irregular Rhythm (a-fib, a-flutter, MAT)

- **Metoprolol** 5mg slow IVP over 2 mins may repeat in 5 mins OR
- **Amiodarone** 150 mg IV over 10 minutes